Souparno Nag

India | +91 8240204932 | souparnonag2004@gmail.com | in.linkedin.com/in/souparno-nag-10994228b | github.com/souparno-nag

Education

Vellore Institute of Technology CGPA: 9.37
Bachelor of Technology in Computer Science *Aug. 2023 – Present*

Technical Skills

Languages: Go, Java, Python, JavaScript, TypeScript, C/C++, SQL, Bash, YAML
Frameworks: Spring Boot, FastAPI, Flask, Django, React, Next.js, Tailwind CSS, LangChain, LangGraph, CrewAI, Rasa
Databases: PostgreSQL, MySQL, Oracle DB, SQLite, MongoDB, Redis, Snowflake, Firebase, Supabase
Developer Tools & Platforms: Git, Docker, Kubernetes, Kafka, Hadoop, ZooKeeper, AWS, Google Cloud, Prisma, Mongoose, Spring JPA
Libraries: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, OpenCV, NLTK, Matplotlib, Seaborn, Tesseract, Qiskit

Experience

Software Engineering Intern June 2026 – August 2026
NatWest Group *Bangalore, India*

- Developing a voice-enabled layer for an internal AI application built with Spring AI, enabling natural language interactions with enterprise banking systems.
- Researching Small Language Models (SLMs) and federated learning for privacy-preserving, on-premise AI for HR & People workflows.
- Creating a central CI/CD repository for GitLab Actions templates across multiple languages, standardizing build and test pipelines.

Frontend and UI Development Internship December 2025 – February 2026
Mirai *Chennai, India*

- Developed the responsive user interface for a company MVP using Next.js, ensuring optimal user experience across all devices.
- Optimized application performance by migrating image storage from base-64 encoded strings to blob storage, significantly improving load times.
- Led full-stack integration with the FastAPI backend to enable dynamic data flow.

Software Development August 2025 – December 2025
Vellore Institute of Technology *Chennai, India*

- Built VBOT, a Retrieval-Augmented Generation (RAG) chatbot for answering institutional queries using official documents.
- Developed end-to-end pipeline including document ingestion, semantic chunking, embedding, and vector retrieval.
- Implemented conversational AI system using LangChain with context-aware memory and prompt engineering.
- Integrated Qdrant/Chroma vector databases with OpenAI embeddings for efficient semantic search and retrieval.
- Built interactive chatbot interface using Streamlit with session-based chat history management.
- Evaluated chatbot using RAGAS, achieving high context precision (0.8) and recall (0.85) for document retrieval.

Summer Research Internship May 2025 – September 2025
Vellore Institute of Technology *Chennai, India*

- Worked on a real time breast cancer dataset, performing data cleaning and feature extraction to make the data useful.
- Applied existing machine learning algorithms on the final data to evaluate their performance.
- Came up an ensemble algorithm that produced higher accuracy than existing ML algorithms.

Projects

DistriCache | *Go, Consistent Hashing, Concurrency*

- Built a distributed in-memory cache in Go using consistent hashing with virtual nodes, achieving under 10% key remapping on node scaling events versus 90%+ with naive modulo sharding.
- Implemented concurrent-safe sharded key-value stores using per-shard `sync.RWMutex` locking to eliminate contention under high-throughput concurrent reads/writes.
- Designed automatic rebalancing on node join/leave, migrating only the affected key range on the hash ring rather than a full reshard.
- Added TTL-based eviction via a background sweep goroutine and replica fallback across adjacent ring nodes for fault tolerance.
- Load-tested throughput and remapping ratios with Go benchmarks to validate performance under concurrent multi-node scaling scenarios.

Event-Driven Ticket Marketplace – Distributed Saga & Real-Time Inventory | *Spring Boot, Apache Kafka, Redis, Res*

- Architected a microservices ecosystem (API Gateway, Auth, Order, Payment, Inventory) with choreography-based Saga orchestrated via Kafka, ensuring eventual consistency across distributed transactions without 2PC.
- Implemented real-time seat locking using Redis Lua scripts with optimistic concurrency control, achieving idempotent booking operations and preventing double-booking under 1k+ concurrent requests.
- Built a CQRS event-sourcing pipeline – raw domain events published to Kafka, consumed by a Spring Boot projection service, and materialized into Elasticsearch for read-optimized seat availability queries.
- Integrated Resilience4j retry, circuit breaker, and time limiter with Spring Cloud CircuitBreaker; deployed on Kubernetes with Helm, Eureka service discovery, Zipkin distributed tracing, and Prometheus metrics.

VonageConnect | *NodeJS, ExpressJS, MongoDB, Vonage API, Rasa Conversational AI*

- Developed a Node.js/Express webhook system integrating SMS and WhatsApp messaging via Vonage Messages API with MongoDB persistence.
- Implemented message handlers, user management controllers, and webhook endpoints for inbound/outbound message processing.
- Built with Mongoose ODM for data modeling and included separate Rasa chatbot component with NLU capabilities.
- Features dual authentication modes, comprehensive message tracking via UUIDs, and supports real-time status updates across multiple messaging channels.

Unified Data Platform – Transaction to Analytics | *PostgreSQL, Kafka, Hadoop, Snowflake, Spark, Airflow*

- Designed an end-to-end lambda architecture pipeline ingesting real-time CDC events from PostgreSQL via Debezium into Kafka, enabling sub-second data availability across downstream consumers.
- Built a stream processing layer using Kafka Streams to enrich and join order and product events, delivering aggregated metrics to Snowflake via Snowpipe for real-time BI dashboards.
- Implemented a Hadoop HDFS data lake storing raw events in partitioned Parquet format, with Apache Spark batch jobs for nightly historical aggregations, data quality checks, and metric recomputation.
- Orchestrated the full pipeline using Apache Airflow DAGs covering Debezium connector management, Spark job triggers, and Snowflake COPY INTO loads with automated retry and failure alerting.

AI Assessment Creator | *Next.js, Node.js, MongoDB, Redis, BullMQ, WebSockets, OpenRouter, Jest*

- Built a full-stack platform where teachers generate structured question papers via AI – cuts manual prep time by 95% using asynchronous job queues (BullMQ + Redis) and real-time WebSocket updates.
- Integrated OpenRouter SDK as a vendor-agnostic LLM gateway (GPT-4o, Claude, Llama) with automatic model fallback, increasing reliability by 40% during peak loads.
- Designed dynamic assignment form with drag-and-drop file upload (PDF text extraction), per-question-type marks, and Zod validation; rendered mobile-responsive exam papers with difficulty badges and PDF export.
- Implemented comprehensive testing (Jest, Supertest, mongodb-memory-server, llmock) and Redis caching, reducing database load by 60% and ensuring deterministic AI mocking.

Serverless E-Commerce Platform on AWS | *AWS (S3, Lambda, API Gateway, DynamoDB, SQS, SES, CloudFront, Cognit*

- Designed and deployed a serverless, event-driven e-commerce platform using 10+ AWS services within Free Tier.
- Built scalable architecture using API Gateway + Lambda for backend logic and DynamoDB for high-performance NoSQL storage.
- Implemented user authentication with Cognito and secure API access using JWT-based authorization.

- Developed asynchronous order processing pipeline using SQS + Lambda, improving system responsiveness and fault tolerance.
- Integrated SES for automated email notifications and CloudWatch for monitoring and logging.
- Achieved high availability, scalability, and zero server management through cloud-native architecture.

RiskPilot | *Python, LangGraph, LangChain, Pandas, NumPy, Streamlit, OpenAI/Claude API*

- Built a multi-agent risk reporting system using LangGraph that orchestrates data retrieval, risk computation, anomaly detection, and natural-language report generation in a single automated pipeline.
- Implemented a Risk Calculator agent computing historical VaR and rolling volatility on market data pulled via yfinance, feeding results into a downstream anomaly detection node using z-score/isolation forest methods.
- Designed an LLM-powered Commentary Writer agent that synthesizes flagged risk anomalies into desk-style natural-language reports, reducing manual report drafting effort.
- Built an interactive Streamlit dashboard to visualize computed risk metrics alongside auto-generated commentary in real time.

Legal AI Assistant | *Python, LangChain, Google Gemini AI, LangGraph*

- Developed a comprehensive legal document processing platform using Python and LangChain framework with Google's Gemini 2.5 Flash API for intelligent document analysis and question-answering capabilities.
- Implemented RAG (Retrieval-Augmented Generation) architecture with Ollama embeddings and vector stores for precise legal document retrieval and contextual question answering with persistent conversation state.
- Built multi-level document summarization system using LangGraph state management with expertise-based output modes (expert, moderate, beginner) optimized for different legal knowledge levels.
- Integrated advanced PDF processing with PyPDF2, recursive text splitting, and asynchronous document handling to support large legal document workflows with token-aware content management and parallel processing.

CreditPulse | *Python, PostgreSQL, Scikit-learn, XGBoost, SHAP, Matplotlib*

- Developed a credit default prediction model (Logistic Regression, XGBoost) on 150K+ loan records, using SQL-based feature engineering and aggregation queries in PostgreSQL for data preparation.
- Applied SHAP-based explainability analysis to identify key drivers of default risk, translating model outputs into interpretable insights for stakeholder review.
- Built an interest rate risk module computing bond portfolio duration and convexity, simulating portfolio value changes under parallel and non-parallel yield curve shock scenarios.
- Visualized rate-shock stress-test results (+100bps/-100bps) to quantify potential P&L impact across the portfolio, supporting risk exposure assessment.

Achievements

- Ranked **1st** in the Electronics and Communications branch during the **1st year** of B.Tech at VIT Chennai.
- Ranked **4th** in the Computer Science branch during the **2nd year** of B.Tech at VIT Chennai.
- Published the research paper *"A hierarchical ensemble machine learning framework for precision survival prediction in breast cancer using SEER data"* in **Results in Engineering (Elsevier)**.